

Group Case 3: Education

Group 8: Team Deadline Daredevils

Aniruddh Maniktala (37332889)

Chion Kim (87525523)

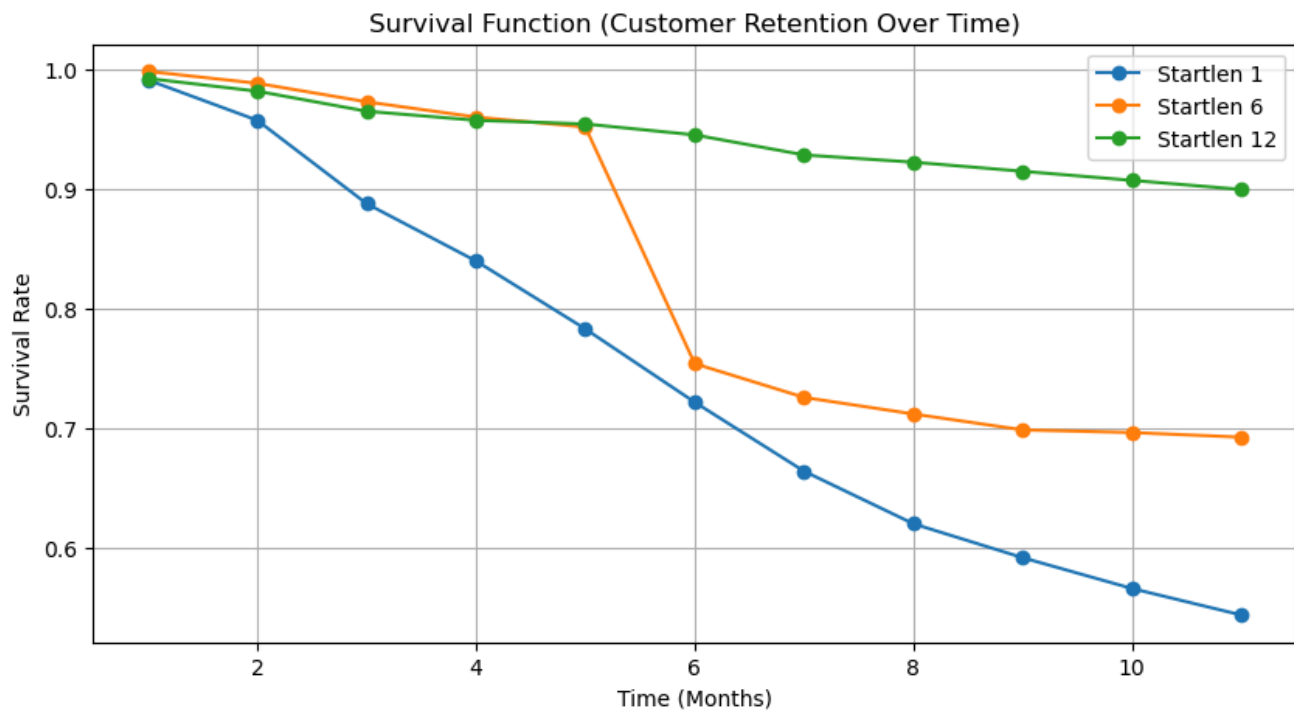
Reshum Zubair (91807925)

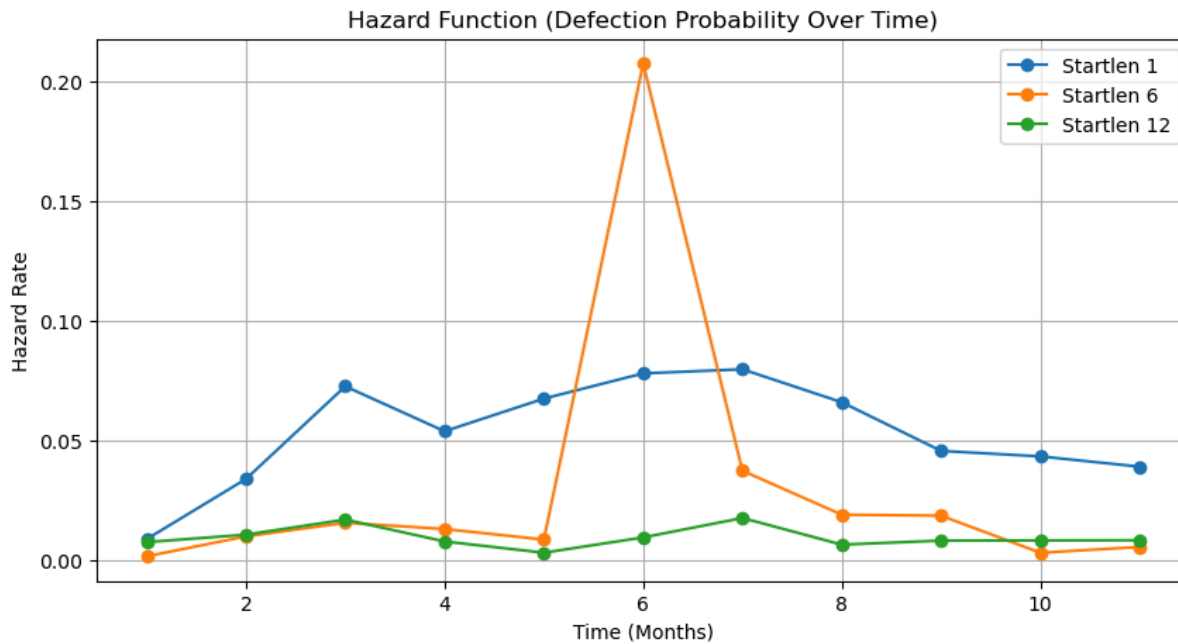
Saloni Gharge (55859177)

Part 1:

Create a chart with the survival function and a chart with the hazard function. Your chart should merely describe the data. That is, do not model the data (i.e., do not impose a functional form), but compute summary statistics that describe the percentage of customers retained (i.e., survival function) and defection probabilities (i.e., hazard function) over time. Plot customers with different levels of startlen separately.

What do you observe? Is the probability of defection constant over time, or does it increase, or decrease? If it is non-constant, is the increase/decrease fairly monotonic and smooth, or are there any spikes?





The defection rate is not constant over time. It varies significantly based on the initial contract length.

1 month contracts

The hazard rate (shown by orange line) is generally high even though there is no major spike. This means that customers on a 1 month contract can churn anytime and it is difficult to predict the cause of churn.

6 month contracts

The hazard rate spikes at 6 months- (by 20%) to its highest point. Meaning that customers churn when their subscription ends and should be offered subscription renewal discounts to maintain their revenue stream. After the peak, the hazard rate stays low, meaning that customers who subscribe are likely to stay.

12 month contracts

These customers are least likely to churn however the risk of churn increases gradually over time. It will be useful to encourage long term contracts.

Part 2:

(1) Estimate a model that includes the original variables as well as the trigger variables.

Optimization terminated successfully.

Current function value: 0.114726

Iterations 10

Logit Regression Results

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Dep. Variable:      cancelnow    No. Observations:      43442
Model:              Logit        Df Residuals:           43425
Method:              MLE         Df Model:              16
Date:                Sun, 23 Mar 2025    Pseudo R-squ.:         0.2892
Time:                21:00:20    Log-Likelihood:        -4983.9
converged:           True         LL-Null:               -7011.6
Covariance Type:     nonrobust    LLR p-value:           0.000
=====

```

	coef	std err	z	P> z	[0.025	0.975]
Intercept	-5.6397	0.255	-22.120	0.000	-6.139	-5.140
C(time)[T.2]	1.1592	0.218	5.312	0.000	0.731	1.587
C(time)[T.3]	1.8853	0.207	9.090	0.000	1.479	2.292
C(time)[T.4]	1.4040	0.214	6.571	0.000	0.985	1.823
C(time)[T.5]	1.5017	0.214	7.016	0.000	1.082	1.921
C(time)[T.6]	2.3823	0.209	11.423	0.000	1.974	2.791
C(time)[T.7]	2.1439	0.212	10.124	0.000	1.729	2.559
C(time)[T.8]	1.8157	0.219	8.276	0.000	1.386	2.246
C(time)[T.9]	1.5522	0.228	6.805	0.000	1.105	1.999
C(time)[T.10]	1.4121	0.239	5.916	0.000	0.944	1.880
C(time)[T.11]	1.3340	0.245	5.447	0.000	0.854	1.814
payleft	0.0053	0.029	0.179	0.858	-0.052	0.063
lagtest	-0.3316	0.161	-2.064	0.039	-0.647	-0.017
lagtest_prevmonth	-1.7332	0.620	-2.794	0.005	-2.949	-0.518
startlen	0.1219	0.017	7.379	0.000	0.090	0.154
cumlagtest	-0.0234	0.004	-6.059	0.000	-0.031	-0.016
contractup	2.2204	0.157	14.134	0.000	1.913	2.528

(2) Interpret the results from the model. What are the managerial insights?

The following are some key insights on the basis of our model interpretation:

Churn peaks at the time of contract renewal

The log likelihood of churn increases over time and peaks at the 6 month mark. Students who have stayed for 6 months are 2.3823 times more likely to churn as compared to those who have stayed for 1 month. Therefore subscription renewal is a trigger. Churn increases significantly over time from the first to seventh month.

Therefore, the management should offer discounts for students who are nearing the end of their 6 month contract or a renewal discount or a free additional month if they extend their subscription before it expires.

Lagtest_prevmonth

If a student has submitted any work in the previous month then their log odds of churn drop by 1.73. So recent engagement is the strongest factor against churn. Therefore, the students who don't submit any assignments must be traced and should be sent motivational nudges like 'things to do', 'you're falling behind', 'one more assignment to reach your goal' to encourage engagement. The variable lagtest indicates that students who are disengaged this month are more likely to churn now. Therefore, inactivity of students must be closely monitored and chat prompts should be triggered like 'Need help? Lets Chat', "Take a quick quiz to test your knowledge."

Contract expiry is a major churn trigger

Students whose contracts expire (contractup = 2.2204) are much more likely to cancel than those whose contracts are still active. This suggests that the end of a contract is a high risk moment for customer churn.

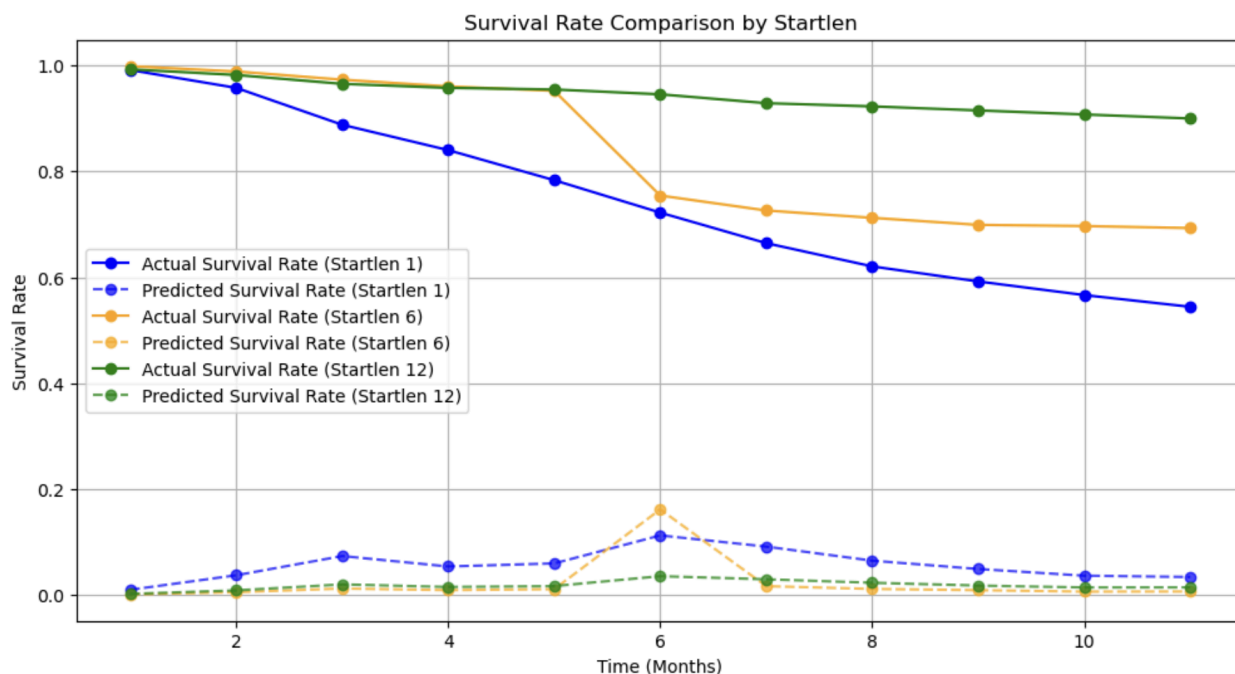
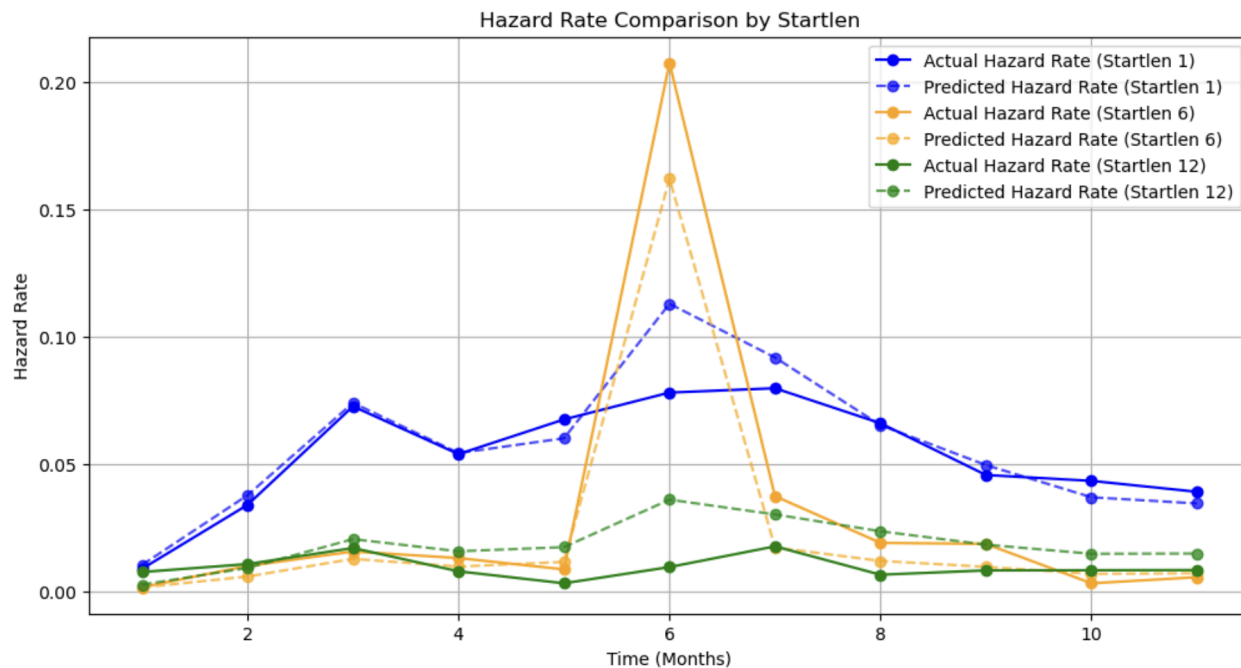
Cumlagtest shows that students who are consistently active over time are more loyal.

Therefore, the company should encourage cumulative participation through streak tracking, badges, or monthly milestones and Highlight students' progress back to them (e.g., "You've completed 10 assignments!")

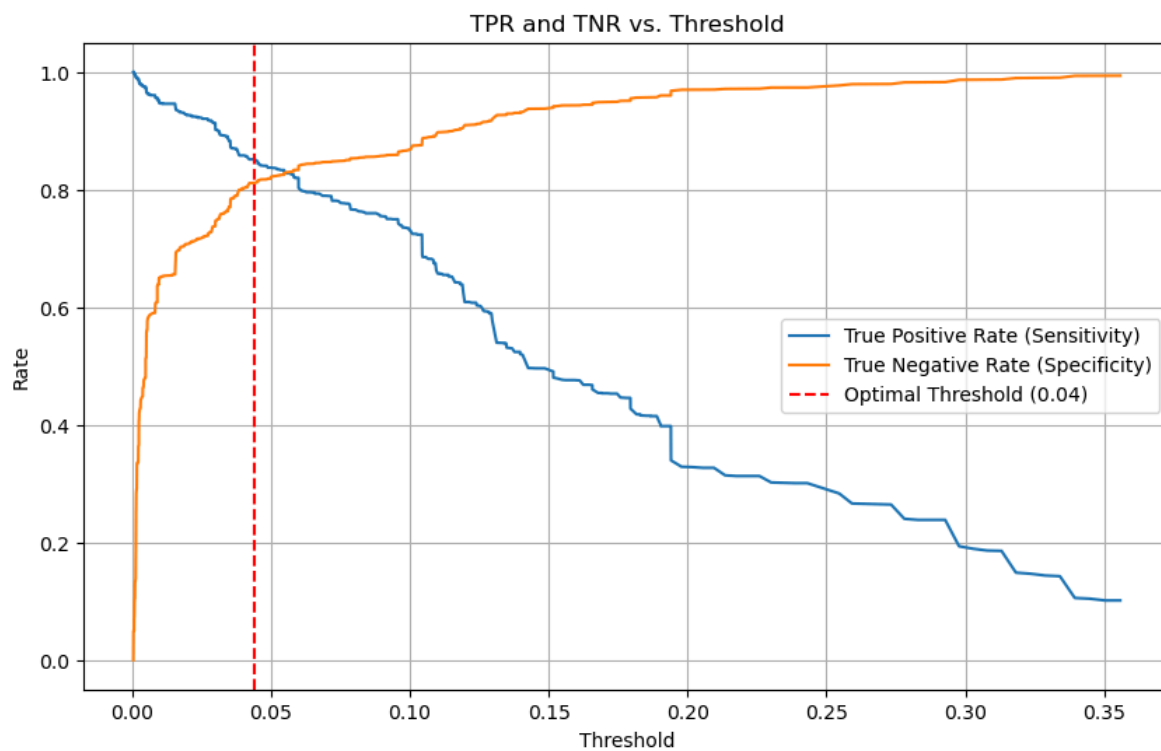
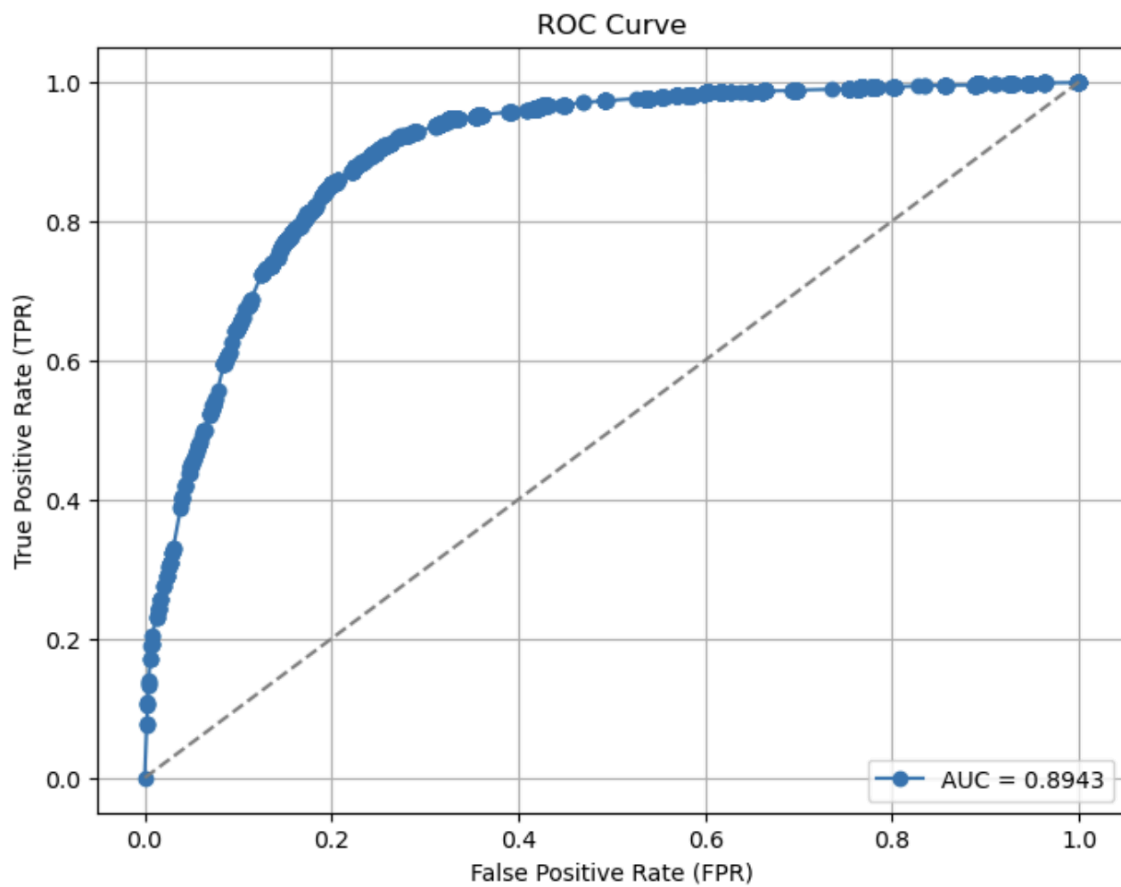
Startlen

The longer the subscription length, lower the odds of cancellation, but it's not that strongly significant. So its not sure that just longer contracts guarantee loyalty. Therefore, the company should not rely on the contract length along and try to sustain engagement throughout the term, especially near the end

(3) Visualize your model-based hazard rate and survival rate.



(4) How does your model perform? Are your model-based predictions close to the raw data?



Our logistic regression model demonstrates a strong ability to identify churn patterns within the dataset. An analysis of the model's predicted hazard and survival rates against the actual raw rates shows a close alignment, particularly during contract renewal periods when churn rates increase.

The ROC curve and AUC score of 0.89 confirm the model's capacity to effectively distinguish between churners and non-churners. The model is proven to perform significantly better than random guessing. The optimal threshold of 0.04 balances Type I and Type II errors and enables this model to achieve an efficient trade-off between maximizing the identification of churners (true positives) and minimizing the misclassification of loyal customers (false positives).

The Managerial Implications of Our Model

Our model prioritizes recall (85%- Model catches most churners) over precision (14%- Few false positives). We have good reason to believe that doing so is in the best interest of the business:

- **Lost Revenue:** If a churner is not flagged ahead of time, the company loses that customer's future revenue, which can be substantial over the course of a subscription year.
- **Customer Lifetime Value:** Missing a churn risk means losing a customer with potentially high lifetime value, which impacts long-term profitability.
- **Intervention Opportunity Cost:** If the company does not intervene in time for a false negative, then there will be no chance to retain that customer. This would result in a permanent loss.

This is not to say that false positives do not carry any cost. Retention efforts or discounts targeting 'safe' customers who are not at risk of churn will unnecessarily add to our costs. However, the cost of these extra interventions is lower compared to the long-term loss from missing a true churner. Therefore, the optimal strategy should lean towards minimizing false negatives, even if it means handling some false positives.